



Groundwater quality assessment using multivariate statistical approach and geospatial modelling around cement industrial corridor, South India

B. Suvarna¹ · V. Sunitha¹ · Y. Sudharshan Reddy¹ · B. Muralidhara Reddy² · A. K. Kadam³ · M. Ramakrishna Reddy²

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Abstract

The current study emphasizes on seasonal hydrogeochemistry, appraisal of groundwater quality for drinking purposes, as well as forms explanation with water quality index studies conducted with geospatial modeling. For this study, total eighty groundwater samples were collected from Yerraguntla, Kadapa district, Andhra Pradesh, during pre-monsoon season of 2019 and post-monsoon season of 2018. The physiochemical examination of groundwater is measured and compared to World Health organization (WHO) standards. The pH ranges were potable decipher. The groundwater is alkaline and hard. Analytical results divulge that the electrical conductivity (EC), total dissolved solids (TDS), total hardness (TH), magnesium (Mg^{2+}), calcium (Ca^{2+}), sodium (Na^+), potassium (K^+), carbonates (CO_3^{2-}), bicarbonates (HCO_3^-), chlorides (Cl^-), sulfates (SO_4^{2-}), nitrates (NO_3^-), and fluorides (F^-) of groundwater samples were exceeding the standards of WHO. Correlation and R-mode factor results are offered to investigate the parameters that control groundwater chemical changes. The most essential parameters, according to the R mode factor analysis, are EC, TDS, TH, Mg^{2+} , Na^+ , Ca^{2+} , Cl^- , HCO_3^- , NO_3^- , and F^- . During the pre- and post-monsoon seasons, most groundwater samples drop into the poor to very poor water class for drinking practices, as per the Water Quality Index and the geochemistry is governed by water–rock interaction and extensive chemical weathering.

Keywords Groundwater quality · Hydrogeochemistry · Water quality index (WQI) · Geographic information system (GIS) · Limestone terrain · Kadapa · India

Introduction

Water is one of the most vital nutrients in the ecosystem. There are living things on earth because this is the only planet that has water. This is essential as long as survival of plants, animals and all living things in this ecosystem. It is the most abundant commodity in nature and the most abused commodity (Suvarna et al. 2018; Adimalla and Venkatayogi

2018; Sunitha et al. 2014a, b). Water quality barriers are particularly disturbed by human intervention and environmental changes, which are increasing the ecological risks of many normal water areas (such as streams, lakes, and coastal lagoons.). Since, water quality observation and monitoring are rooted in the preservation and conservation of nature as well as the earth's future changes. According to previous literature, groundwater contamination is related to several sources, such as geogenic, anthropogenic, nitrogen pollution, and the dissolution of rocks and soil in agricultural practices (Sudharshan Reddy et al. 2020). The chemical properties of groundwater are primarily determined by the composition of the matrix rock as a result of geological activity. Highly limited factors such as topography and lithology affect the quality to change within a short distance of the inspected area (Raju et al. 2010; Suvarna et al. 2019a, b). The chemical properties of groundwater can be changed through different anthropogenic practices. These anthropogenic sources like point sources such as waste treatment practices and

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✉ V. Sunitha
vangalasunitha@gmail.com

¹ Department of Geology, Yogi Vemana University, Kadapa, Andhra Pradesh, India

² Department of Earth Sciences, Yogi Vemana University, Kadapa, India

³ Department of Environmental Sciences, Savitribai Phule Pune University, Pune, India



sanitary conditions (Subbarao et al. 2012). This amount is also affected by weathering, groundwater movement, ion exchange mechanisms, climate change, and fluctuations in replenishment and emissions. The geochemistry of water is controlled by the ensuing factors: (i) geochemistry of rocks and soil (ii) monsoon and semi-arid climate with water shortage in summer (iii) exchange between aquifers due to pressure levels caused by excessive extraction (iv) groundwater contaminated by contaminated surface water (v) sewage goes directly into poorly designed wells and (vi) the extent of water use. The chemical composition of minerals or rocks is affected by the elemental composition in the form of main elements or trace elements (Karunanidhi et al. 2019; Adimalla and Qian et al. 2019). For clastic rocks of

sedimentary origin, the genetic history can be determined by determining the grain size, structure, roundness and sphericity, while chemical analysis of nonclastic rocks (such as argillaceous and calcareous sedimentary rocks) is of diagnostic value (Arumugam and Elangovan 2009; Abdulsoul et al. 2015; Subbarao 2018). This may give consideration to the nature of the source material, transportation process and deposition environment set up Tables 1 and 2.

Water scarcity has emerged as a significant universal problem concern for long-term evolution. As previously mentioned, it has become more critical as the world's inhabitants continue to grow, resulting in, for example, a growth in the quality of living, changes in dietary habits, and anthropogenic influence on climate change. Groundwater is a major

Table 1 Statistical summary of physico chemical parameters of groundwater for PRM and POM seasons ($n = 40$)

Chemical parameter	Expressed	PRM			POM		
		Minimum	Maximum	Mean	Minimum	Maximum	Mean
pH	Units	7.3	9.2	8	7.1	8.3	7.7
EC	$\mu\text{S/cm}$	1400	9865	4880	1380	9700	4753
TDS	mg/L	680	7800	2474	660	7700	2397
TH	mg/L	120	1650	530	100	1600	505
Na^+	mg/L	24.4	570	240	22	560	227
K^+	mg/L	0.34	92.3	25.5	0.2	90	24
Ca^{2+}	mg/L	25	350	93.2	24	344	85.4
Mg^{2+}	mg/L	3.24	310	105	3.4	310	106
CO_3^{2-}	mg/L	0	340	99	0	300	89
HCO_3^-	mg/L	180	1340	570	170	1244	547
SO_4^{2-}	mg/L	48	1400	394	45	1280	374
Cl^-	mg/L	70	1820	504	68	1775	474
NO_3^-	mg/L	2.5	760	86.1	2.2	751	82.3
F^-	mg/L	0.45	6.24	1.7	0.4	5.8	1.5

Table 2 Comparison of the quality parameters of groundwater samples of the study region with WHO standards for PRM and POM seasons

Chemical parameter	Expressed	Ranges of WHO (2011) standards		% Samples exceeding WHO limits			
				PRM		POM	
		Desirable	Permissible	Desirable (%)	Permissible	Desirable (%)	Permissible
Ph	Units	7	9.2	100	Nil	100	Nil
EC	$\mu\text{S/cm}$	1500	1500	2.5	97.5%	2.5	97.5%
TDS	mg/L	500	1500	100	67.50%	100	67.50%
TH	mg/L	100	500	100	45%	100	42.50%
Na^+	mg/L	50	200	98	60%	98	62.50%
K^+	mg/L	10	10	57.5	42.50%	57.5	42.50%
Ca^{2+}	mg/L	75	200	52.50	7.50%	45	7.50%
Mg^{2+}	mg/L	30	150	95	22.50%	95	22.50%
HCO_3^-	mg/L	300	600	87.50	37.50%	85	35%
SO_4^{2-}	mg/L	200	600	65	15%	62.50	15%
Cl^-	mg/L	250	600	65	35%	65	32.50%
NO_3^-	mg/L	45	45	42.50	42.50%	42.50	42.50%
F^-	mg/L	0.5	1.5	97.50	45%	97.50	40%



source of potable water in both rural and urban provinces. Aquifers all over the world are facing an increased threat of contamination due to industrial expansion, urbanization, and other factors. As a result, a comprehensive research of groundwater quality is required, which will consequence in hands-on campaigns and practical initiatives targeted at preserving groundwater's natural condition. It is essential for domestic, agricultural, and industrial utilize (Babiker et al. 2007).

Anthropogenic activities include industrial production and waste disposal, as well as the indiscriminate release of several heavy metals into the ecosystem. Due to the increased densities of industries across a small region, water contamination by industrial operations is a common occurrence in the surrounding communities (Bhutiani et al. 2017). In general, industrial activities are among the leading causes of environmental damage. Metal mining, smelting, and meat processing, as well as oil and gas exploration, petrochemical manufacture, and different industrial operations, have a significant negative impact on the ecology. These negative environmental effects are caused by direct industrial activities, waste releases, and products that are often poisonous in their untreated stages.

Cement plants are known to emit volatile organic compounds and micro-pollutants. Soil, ambient air, and surface water in the area of a cement factory have all been shown to be severely damaged, according to studies (Siddiqui et al. 2011). Degraded runoff from contaminated particulate matter in the air, cooling water, leachates from stockpiles of raw materials, and effluents from quality control laboratories have all been connected to stream and river pollution. In the present study region of Yerraguntla, open cast and underground limestone mines and cement industries are present; several tons of waste material are removed and dumped near to the mine premises (Suvarna et al. 2020). These artificial barren hills of lime stone waste dumps exposed to wind and water produce dispersion of metal through natural transporting media. Dispersion of metals not only depends on mining activities, but also on the physico-chemical conditions of the deposits, leaching contaminants from tailing ponds and slack piles are also source of groundwater contamination from mining operations. High pH, EC, TDS, and TH are all features of mine drainage wastes. Therefore, in the present study area mine drainage waste is also one of the contributing factors for high pH, EC, TDS, TH in the groundwater (Ipeaiyeda and Obaje 2017).

Although the uniformity of the chemical composition is expected to occur in a wide range, due to the uniformity and small fluctuations of the deposition environment conditions,

these circumstances are generally distant from ideal and will change in time and space, which affects the manifestation of the chemical composition. Horton (1965) basically created WQI in the USA and is widely applied in Africa and Asia (Handa 1985; Mohan et al. 2000; Li et al. 2018). Various researchers have attempted to develop a variety of water quality indicators for assessing groundwater quality. The choice of indicators depends on the input parameters of the groundwater and the desired results (Sunitha and Reddy 2022; Muralidhara Reddy et al. 2016, 2019). For instance, the WQI is an effective technique that can be applied to decipher the suitability of drinking water quality in any area and convey overall water quality information.

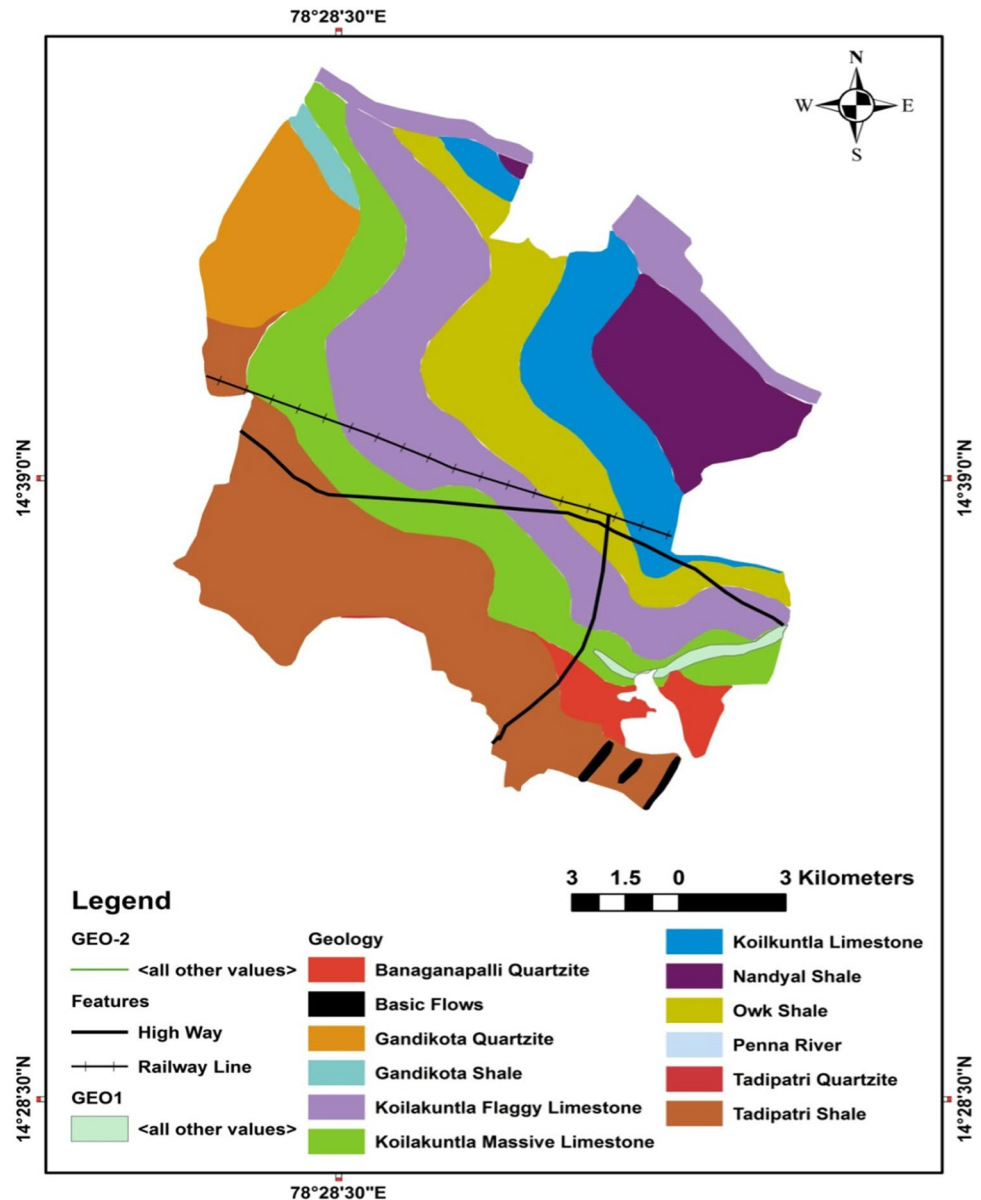
Groundwater currently exists the prime source of drinking water in several rural and suburban provinces, as well as certain urban areas. As a result, a vast population is at risk of ingesting tainted water. The environment has a significant effect on human health and evolution (Zacchaeus et al. 2020; Temilola et al. 2011). Noticing environmental hazard time trends can be helpful to understand the back-drop and contamination. Geochemistry of groundwater and its sustainability for drinking purpose has not yet studied in this region, and hence, this study was emphasized on an assessment of suitability of water for drinking practices. Groundwater is the important origin of drinking water and cultivation needs in the semi-arid area of Yerraguntla, Y.S.R Kadapa District, South India. Hence, the principal purpose of this region intend to determine the drinking water suitability through computation of water quality index (WQI) method and Geographic information system (GIS) applications for understanding the quality dispersal in the study region. This work also focuses on classifying the geochemistry and hydrogeochemistry in and around cement industrial corridor of Yerraguntla region, South India, during pre-monsoon (PRM) season of 2019 and post-monsoon season (POM) of 2018.

Study area

The present study location is situated in between Latitudes $N14^{\circ}35'17.2''$ – $N14^{\circ}44'38.4''$ and Longitudes $E78^{\circ}27'38.4''$ – $E78^{\circ}34'39.3''$ and is present in toposheet No 57 J/06 and J/10 (Fig. 1). The present study region belongs to the formations of the Kurnool Group rest over the rocks of the Cuddapah Super group and the granitic basement unconformably. In fact, there is an overlap of the Narji Limestone on to the granitic basement (Fig. 1) (Nagaraja Rao et al. 1987). The presence of bedding planes, joints, fractures, fissures, and



Fig. 1 Geology map of Yerraguntla, Kadapa, A.P

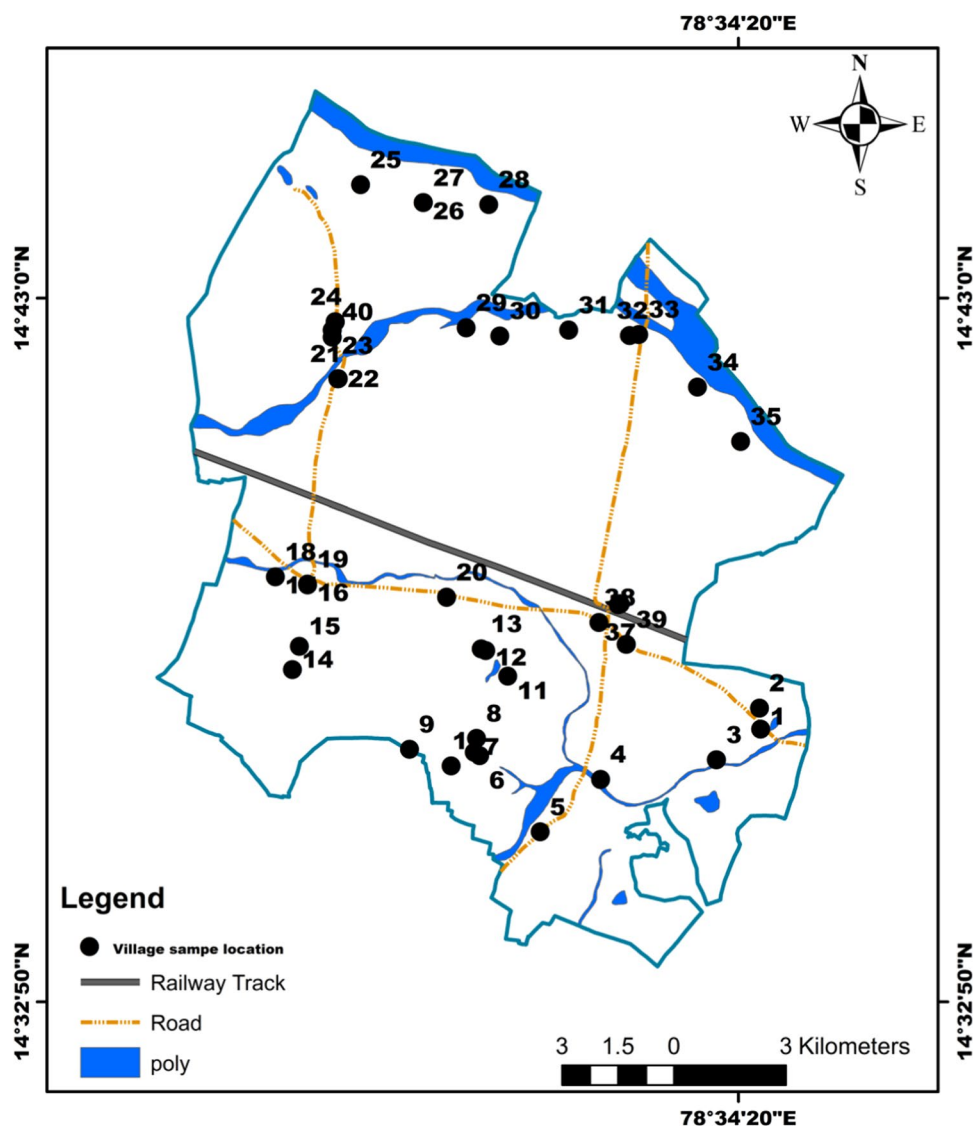


solution voids in this group of rocks controls groundwater. Dolomitic limestones have a network of fractures and joints that aid water percolation and the creation of phreatic aquifers. In limestone locations, karst springs (cave springs) are also found, resulting in enormous amounts of water (CGWB 2016). Bore wells are typically 40 to 70 m deep. The bore well yields range from 1100 to 18,000 lph. Pennar is the main River flowing in NW–SE direction. Sub-dendritic to parallel drainage

pattern is noticed in this region. Mean annual rainfall of this area is 730.7 mm (Ramakrishna Reddy et al. 1998; CGWB 2013; Suvarna et al. 2019a, b). Structural landforms, denudational and fluvial forms are important geo-morphological features noticed in this region (Suvarna et al. 2019a). It is noted that black soil covers 80% of the land. Cotton and groundnut are often grown in study region.



Fig. 2 Sample location map of the study region



Materials and methods

Sampling and analytical procedure

Groundwater samples were obtained from 40 bore wells covering 20 different villages in and around Yerraguntla Mandal, Y.S.R Kadapa district, Andhra Pradesh, during PRM season of 2019 and POM season of 2018. The sampling stations were picked up at random depending on the existence of bore wells in addition to hand pumps in the surrounding area. A portable GPS device was used to record sampling locations, which currently are depicted in Fig. 2. Flowchart adopted by the methodology is shown in Fig. 3. Samples that were collected in pre-cleaned in addition to fully dried

polyethylene bottles were carried to the laboratory. pH, electrical conductivity, total dissolved solids were resolved by pH meter, EC meter, TDS meter, respectively (Raghunath 2003) and Total Hardness. Major cations like calcium, magnesium were measured by titrimetry, whereas Na^+ and K^+ are resolved by flame photometer (Systronics Model No:128). Major anions like carbonates, bicarbonates, chlorides were measured by titrimetry; sulfates and nitrates are analyzed by spectrophotometric method. Ion selective electrodes are used to determine fluoride (Orion 4-star ion meter, Model: pH/ISE). Standard fluoride solutions (0.1–10 mg/L) were made from a sodium fluoride stock solution (100 mg/L). 2 ml of total ionic strength adjusting buffer grade III (TISAB III) was added to 20 ml of material to meet the experimental requirements (Sunitha and Muralidhara 2014; Hem 1985; Raghunath, 1987; APHA 2012).



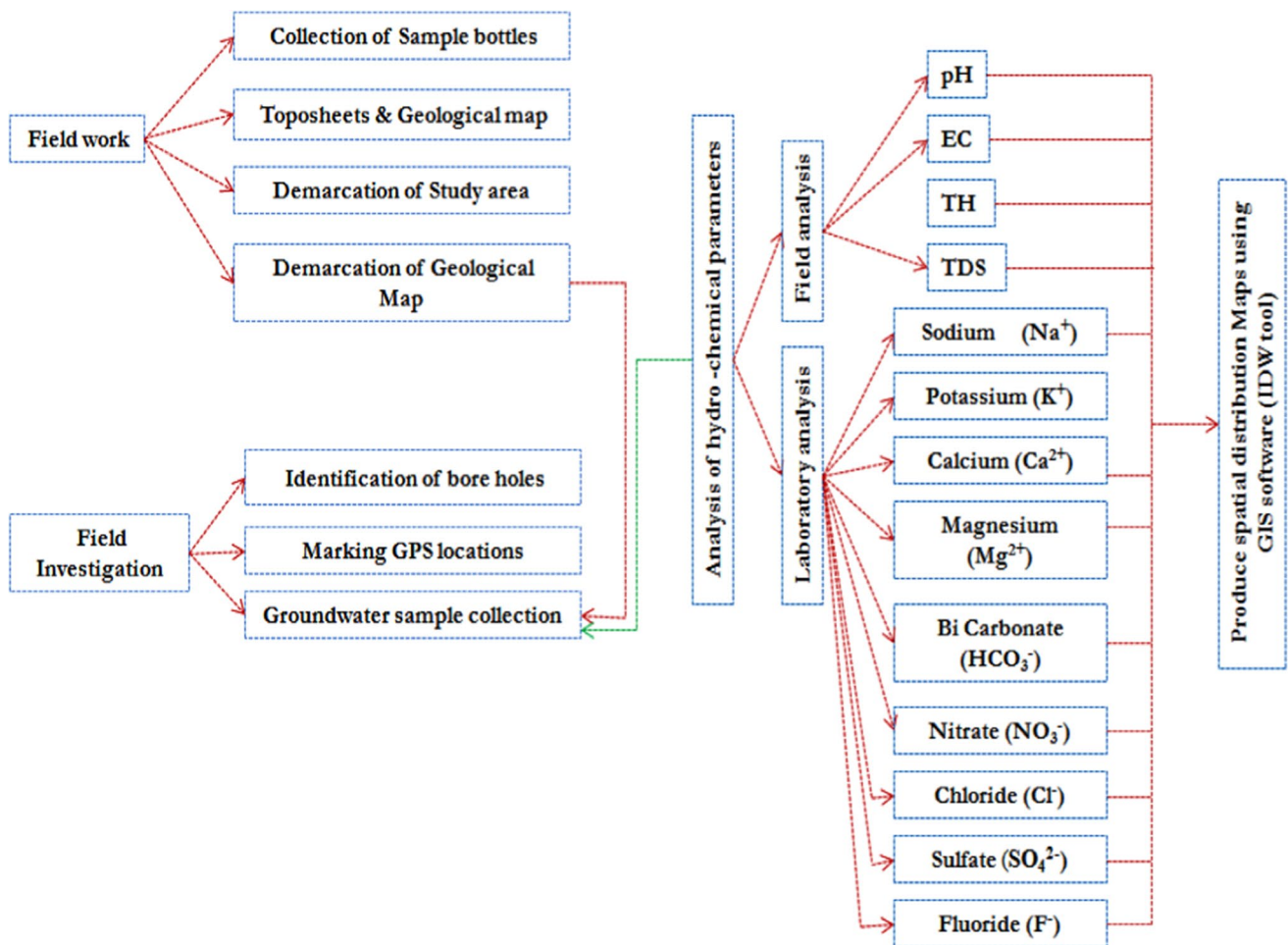


Fig. 3 Flowchart adopted for methodology

Analytical accuracy

Ionic balance error (IBE) was verified to confirm precision among the concentration of total cations (Na^+ , K^+ , Ca^{2+} , Mg^{2+}) and the total anionic concentrations of (HCO_3^- , Cl^- , SO_4^{2-} , NO_3^- and F^-) depicted in milli equivalent per litre (meq/L) for all samples using the equation below:

$$\text{IBE} = \frac{\sum \text{Cations} - \sum \text{Anions}}{\sum \text{Cations} + \sum \text{Anions}} \times 100 \left(\frac{\text{meq}}{\text{L}} \right)$$

IBE value is less than the admissible level of ± 5 (Domenico and Schwartz 1990).

Multivariate statistical analysis

The water quality data sets were submitted to multivariate analysis utilizing Pearson correlation matrix and factor analytical approaches. The Statistical Package for Social

Sciences (SPSS-64) software was applied to calculate the above the statistical parameters. Combining hydrogeochemical analysis with geographic information systems (GIS) can help describe the applicability of groundwater quality for several uses (Muralidhara Reddy et al. 2013, 2015). Arc GIS 10.2 (spatial analyzer module) produces groundwater quality maps in an easy way to understand the format. The weight is a function of inverse distance in this method, and similar points had greater influence in determining unknown values (Burrough and Mc Donnell 1998).

Water quality index (WQI) estimation

In addition, water quality index was computed by applying a weighted arithmetic index method. WQI used was developed by Horton (1965). The determination of WQI the following three steps was applied: (1) water quality rating (q_n) calculation (depending on Eq. (1)); (2) calculation for Unit weight



(W_n) (depending on Eq. 2); and (3) the WQI is calculated by Eq. (3):

$$q_n = \left[\frac{V_n - V_i}{S_n - V_i} \right] \times 100 \quad (1)$$

where q_n is the water quality rating for the n th parameter, V_n is the noted value for the n th parameter, S_n is the suggested Standard for the corresponding parameter, and V_i is the ideal value for the n th parameter. Excluding for pH 7.0, all ideal values (V_i) for drinking water are set to zero.

$$W_n = \frac{k}{S_n} \quad (2)$$

where W_n is the unit weight for the n th parameter, and K is the proportionality constant. By linearly aggregating the quality rating with the unit weight, the overall water quality index was determined.

$$WQI = \frac{\sum q_n W_n}{\sum W_n} \quad (3)$$

Results and discussion

pH, electrical conductivity (EC), total dissolved solids (TDS), total hardness (TH)

The pH of groundwater water sample ranges from 7.31 to 9.2 with an average of 8.0 during PRM season and from 7.1 to 8.31 with a mean value of 7.7 during the POM season. All of the groundwater samples are of alkaline in nature. To know the deviation in pH, it is grouped into four class (1) 6.5–7.0 (2) 7.0–7.5 (3) 7.5–8.0 (4) 8.0–8.5 as shown in Table 3. The main origin of pH in groundwater is carbonates and bicarbonates. The modest increase in pH during the PRM season can be due to the lack of CO_2 availability (Subbarao et al. 2017). Despite the fact that cement effluent is alkaline, the downstream water pH barely increased by 0.1 units as compared to the upstream water pH in this investigation. The pH

of the water in this area ranged between 6.5 and 9.0, which is ideal for drinking water and aquatic life (Table 3).

Conductivity is a measurement of electrical conduction that demonstrates the ability of materials to pass a current. The significant preferable level of EC in drinking water is denoted as 1500 $\mu S/cm$ (WHO 2011). The conductivity of groundwater sample ranged from 1400 to 9865 $\mu S/cm$ with a mean of 4880.3 $\mu S/cm$ during the PRM season and from 1380 to 9700 $\mu S/cm$ and a mean of 4753.6 $\mu S/cm$ during POM season (Table 1). Higher electrical conductivity can be due to high mineral content; increased salinity at the studied location, temperature, types of ions and moreover electrical conductivity values varies linearly with TDS concentration. 97.5% of the samples are greater than the admissible range of EC. High electrical conductivity may be due to black cotton soil of this area (CGWB 2016). Higher variations in electrical conductivity might also be due to geochemical processes and anthropogenic and industrial pursuits in this area (Zacchaeus et al. 2020; Subbarao et al. 2017; Narasimha and Sudarshan 2013).

The total dissolved solids designate the salinity content of water (USGS 2000; Ranjit Singh et al. 2004). The TDS of groundwater sample ranged from 680 to 7800 mg/L with a mean of 2474.5 mg/L during the PRM season and from 660 to 7700 mg/L with a mean of 2397.8 mg/L during the POM season. TDS are used to determine an average the hardness of water and are used to judge its quality. TDS are a measurement of all salts in solution.

According to WHO (1993) guidelines, 100% samples were present in poor and unsuitable class for drinking purpose during the PRM & POM seasons (Table 4). TDS of groundwater relies on geology, geomorphology, porosity in addition to permeability of rock formations. Water contamination by anthropogenic practices such as agricultural practices, sewage disposal (CGWB 2016). TDS levels beyond a certain threshold have an impact on the water taste, hardness, and corrosive properties (Hari Haran 2002; Subhadra Devi et al. 2003). Though no evidence on the health impacts of ingesting TDS in drinking water appears to exist, many researchers have looked into the relationship among several health effects and hardness, when compared to TDS level (Zacchaeus et al. 2020).

Table 3 pH based classification of groundwater

pH range	PRM		POM	
	Total samples	% of samples	Total samples	% of samples
6.5–7	0	0	0	0
7–7.5	3	7.50%	10	25%
7.5–8	19	47.50%	22	55%
8–8.5	18	45%	8	20%

WHO, World Health Organization;

Table 4 Groundwater classification based on TDS (after WHO 1993)

Total dissolved solids (TDS) mg/L	Water classification	Samples fall (%) in	
		PRM	POM
< 300	Excellent	0	0
300–600	Good	0	0
> 600	Poor and unsuitable	100%	100%



The most prominent attribute of natural water is hardness, which is based mostly on the amount of carbonate, calcium, and magnesium. It is measured in equal CaCO_3 amounts. The total hardness (TH) is the total of temporary and permanent hardness. TH for drinking purposes is acceptable up to 100 mg/L as per WHO international standard (WHO 2011). Groundwater greater than 300 mg/L is of very hard nature (Ritesh Vijay et al. 2010). The TH of groundwater sample ranged from 120 to 1650 mg/L with a mean of 530.5 mg/L during the PRM season and from 100 to 1600 mg/L with a mean value of 505 mg/L during the POM season. Sawyer and McCarty (1967) classification depicts that 70% and 67.5% of the samples belong to very hard water class, while 25% and 27.5% of samples fall under hard water category during the PRM and POM seasons (Table 5).

The high degree of hardness in the current region could be mostly owing to industrial wastes and poor disposal of untreated sewage. Due to the leaching of calcium and magnesium bicarbonate through recharge, pre-monsoon samples have a higher level of hardness than post-monsoon samples (Karanth 1987). There are no adverse effects related to water hardness in the WHO guidelines (WHO 2011); however, Schroeder (1960) mentions the adverse effects of hard water in heart disease and household utilities. However, relation of hard water could be identified as a risk element that can aggravate eczema. The environment is critical in the etiology of atopic eczema; however, the exact cause is unclear (Langan 2009).

Major cation

The sodium of groundwater sample ranged from 24.45 to 570 mg/L with an average of 240.10 mg/L during the PRM season and from 22.32 to 560 mg/L with a mean of 227.51 mg/L during the POM season. 97.5% of the ground water samples are above the sodium desirable limit (50 mg/L) during the PRM and POM seasons. 60% of the groundwater samples are exceeding the sodium allowable limit (200 mg/L) of WHO (2011) guidelines during both seasons (2). Because of the base exchange reaction between clay minerals and zeolite, sodium concentration may be

increased (Karanth 1987). Other causes of sodium contamination include rainwater, dissolving of evaporated minerals, industrial runoff from dispersed sources, and municipal trash discharges (Handa 1985).

The potassium of groundwater sample ranged from 0.342 to 92.32 mg/L with a mean of 25.54 mg/L during the PRM season and from 0.257 to 90.23 mg/L with a mean value of 24.68 mg/L during the POM season. As per WHO (2011) standards, 40% and 37.5% of the groundwater samples are exceeding the potassium allowable level during the PRM and POM seasons, respectively. Because of its increased resistance to chemical weathering and adsorption on clay products, potassium is a major source of groundwater in the current region (Subbarao et al. 2012).

The calcium of groundwater samples ranged from 25 to 350 mg/L with a mean of 93.28 mg/L during the PRM season and from 24.1 to 344 mg/L with a mean value of 85.6 mg/L during the POM season (Table 1). The permissible range of calcium in drinking water is 200 mg/L (WHO 2011). Only 7.5% of groundwater samples are deviating the permissible level during both the seasons (Table 2). Calcium is a major cation and naturally present in the water. In the present region, it might be leached from dissolution of minerals like limestone, marble, calcite, dolomite, gypsum, fluorite, and apatite. The elevated calcium concentration could also be attributed to industrial effluents from industries that use chemicals during their operations (Gaikwad et al. 2020).

The magnesium of groundwater sample ranged from 3.24 to 310.24 mg/L with an average of 105.93 mg/L during the PRM season and from 3.45 to 240.45 mg/L with a mean value of 101.03 mg/L during the POM season (Table 1). As per WHO (2011), 98% of the samples are exceeding the desirable level (30 mg/L) of magnesium during the PRM and POM seasons. 22.5% of the samples are above the permissible limit (150 mg/L) during the both seasons (Table 2). Industrial effluents are also causative factors of high value of magnesium in the groundwater (Zacchaeus et al. 2020; Pondhe et al. 1997; Datta and Tyagi 1996). The higher contents of Mg^{2+} in pre-monsoon suggest that Mg^{2+} is derived from the lithology, i.e., the magnesium deposits present in the alluvium as well as from the limestone and also dissolution of precipitates $\text{Ca,Mg}(\text{CO}_3)_2$ during recharge (Datta and Tyagi 1996; Lakshmanan et al. 2003).

Table 5 Hardness based classification of groundwater (Sawyer and McCarty 1967)

Total hardness (TH) mg/L	Water classification	Samples fall (%) in	
		Pre monsoon	Post monsoon
0–75	Soft water	0	0
75–150	Moderately hard water	5%	5%
150–300	Hard water	25%	27.50%
> 300	Very hard water	70%	67.50%

Major anion

The bicarbonate of groundwater samples ranged from 180 to 1340 mg/L with an average of 570.56 mg/L during the PRM season and from 170 to 1244 mg/L with a mean value of 101.03 mg/L during the POM season. As per WHO and BIS guidelines, 88% and 85% of the samples are exceeding the

standard level (300 mg/L) of bicarbonates during the PRM & POM season, respectively. 37.5% and 35% the samples are exceeding the prescribed level (600 mg/L) of bicarbonates through the PRM and POM seasons. This denotes dissolution of carbonate rocks (limestone, dolomites). The greater concentration of HCO_3^- is due to organic matter breakdown and root respiration, which interacts with rain water (H_2O) to generate HCO_3^- (Eqs. 4 and 5), which regulates pH.



The sulfates of groundwater sample ranged from 48 to 1400 mg/L with a mean of 394.1 mg/L during the PRM season and from 45.2 to 1280.65 mg/L with a mean value of 374.2 mg/L during the POM season. The chief sources of sulfate in groundwater are pyrite crystals, gypsum and anhydrite minerals found in sedimentary rocks are other sources of sulfate in groundwater. According to WHO (2011) guidelines, 65% and 62.5% of the groundwater samples exceeding the desirable range (300 mg/L) of sulfate during the PRM and POM seasons. Sulfate levels may rise in the pre-monsoon season as a result of leaching, anthropogenic and industrial practices in the environment, such as the discharge of industrial waste and urban utilities, which are oxidized and enter the groundwater system (Saxena 2004).

The chloride of groundwater sample ranged from 70.24 to 1820 mg/L with an average of 504.13 mg/L during the PRM season and from 68.02 to 1775.82 mg/L with an average value of 474.89 mg/L during the POM season (Table 1). Groundwater chlorides are higher in the PRM than in the POM season. As per WHO (2011) guidelines, 72.5% and 70% of the groundwater samples are exceeding the desirable range (250 mg/L) of chloride during the PRM & POM seasons, respectively. 35% and 32.5% of the samples are above the permissible range of chloride during the PRM and POM seasons, respectively (Table 2). Higher chlorides

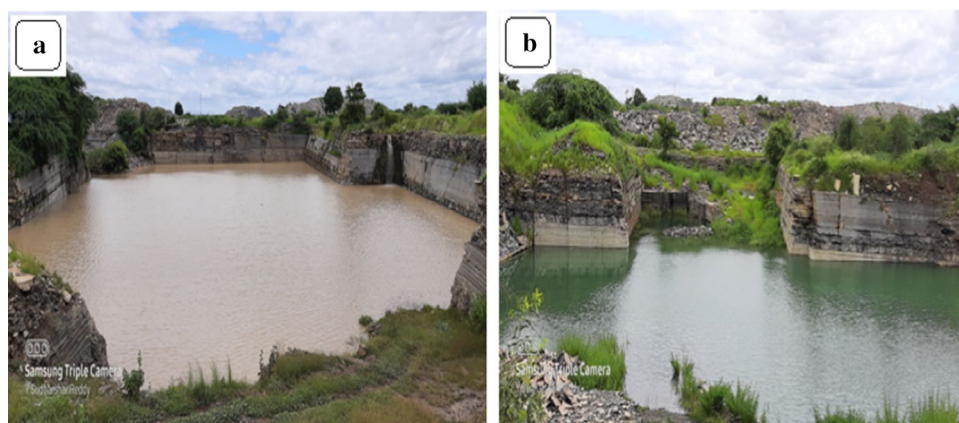
are generally applied as a pollution indicator and as a tracer for groundwater contamination (Loizidou and Kapetanios 1993). Weathering, leaching of rocks and soil, and domestic and municipal effluents are all potential sources of chloride. Higher chloride concentrations are also caused by leaching of upper soil layers by home, municipal, and industrial pollutants, as well as a dry climate (Sunitha and Sudharsan 2019; Ipeaiyeda et al. 2017; Subbarao et al. 2017). Higher chlorides in drinking water have a salty flavor and can cause constipation in persons who are not used to it.

The nitrate of groundwater sample ranged from 2.5 to 760.12 mg/L with an average of 86.13 mg/L during the PRM season and from 2.29 to 751.15 mg/L with a mean value of 82.38 mg/L during the POM season. The important indicators of nitrate in the groundwater are nitrate fertilizers, sewage waste, and decaying organic matter (Karanth 1987; Muralidhara Reddy et al. 2019; Ipeaiyeda et al. 2017; Sunitha et al. 2012, 2022a, b). As per WHO (2011) standards, 45% and 42.5% of the samples are exceeding the prescribed level (45 mg/L) of nitrate during the PRM and POM seasons, respectively (Table 2).

In addition to agricultural land, there are inactive/abandoned mines in surrounding residential neighborhoods in the study region. Because a substantial amount of household and agricultural waste (chemical fertilizers, insecticides, and other chemicals) is deposited in inactive/abandoned mining regions. Increased fertilizer output necessitates the addition of nitrogen and potassium. In this location, nitrate is used, and the potassium fertilizer application rate for the total crop is 200 kg/h every 2–3 weeks (CGWB 2016). Industrial dumping materials, agricultural operations, and septic tanks will all be carried into dormant mineral formations in the current research region, resulting in greater nitrate concentrations (Zacchaeus et al. 2020; Ipeaiyeda et al. 2017) (Fig. 4).

The fluoride of groundwater sample ranged from 0.45 to 6.24 mg/L with a mean of 1.77 mg/L during the PRM season and from 0.49 to 5.81 mg/L with a mean value of

Fig. 4 Nitrate contamination sources in study area showing stagnated water from an abandoned mine



1.52 mg/L during the POM season, respectively (Table 1). As per WHO (2011) guidelines, 45% and 40% of the samples are exceeding the prescribed level (1.5 mg/L) of fluoride during PRM and POM seasons, respectively (Table 2). The principal sources of fluoride in groundwater are geogenic processes. The high fluoride range in this area's groundwater could be attributed to fundamental mobility and the production of sedimentary rocks including limestone, shale, and carbonates (Brindha et al. 2011; Brindha and Elango 2013). As a result of the ion exchange reaction, water–rock interaction, and other factors impact mineral solubility in the research region. The presence of clay in the research area is the primary source of F⁻ contamination in the groundwater (Rao et al. 2014; Subbarao 2017). A secondary source of enhanced F⁻ in groundwater is the widespread use of phosphate fertilizers to boost crop yields (Subbarao et al. 2018).

Multivariate statistical analysis

Correlation matrix

In the present study region, Pearson's correlation analysis of PRM and POM seasons is shown in Table 6. EC has strong correlation with TDS, TH, Mg²⁺ and moderate relation with Na⁺, Ca²⁺ suggesting cessation mechanisms which changes the salt content (Wagh et al. 2016). TDS is significantly correlated with TH and Mg²⁺, but only a minor correlation with Ca²⁺ and Na⁺, indicating that ions that contribute to TDS are dissolving. The dissolved salts of Ca²⁺ and Mg²⁺ alter the TH of groundwater. This supports the theory that these ions are produced through ion exchange silicate weathering materials. (HCO₃⁻ + SO₄²⁻) versus (Ca²⁺ + Mg²⁺) scatter diagram

Table 6 Correlation matrix between chemical parameters

		pH	EC	TDS	TH	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	CO ₃ ²⁻	HCO ₃ ⁻	SO ₄ ²⁻	Cl ⁻	NO ₃ ⁻	F ⁻
pH	Pre	1.000													
	Post														
EC	Pre	-0.287	1.000												
	Post	-0.425													
TDS	Pre	-0.226	0.943	1.000											
	Post	-0.387	0.947												
TH	Pre	-0.203	0.761	0.822	1.000										
	Post	-0.476	0.747	0.805											
Na ⁺	Pre	-0.173	0.670	0.694	0.433	1.000									
	Post	-0.221	0.677	0.726	0.446										
K ⁺	Pre	-0.060	0.226	0.288	0.201	0.487	1.000								
	Post	-0.130	0.173	0.295	0.210	0.476									
Ca ²⁺	Pre	0.075	0.535	0.699	0.630	0.490	0.217	1.000							
	Post	-0.080	0.514	0.674	0.627	0.470	0.215								
Mg ²⁺	Pre	-0.251	0.720	0.758	0.981	0.386	0.194	0.488	1.000						
	Post	-0.510	0.731	0.745	0.982	0.391	0.181	0.481							
CO ₃ ²⁻	Pre	-0.087	-0.025	-0.020	0.070	0.157	0.195	-0.163	0.104	1.000					
	Post	-0.038	-0.003	-0.004	0.085	0.174	0.185	-0.152	0.139						
HCO ₃ ⁻	Pre	-0.125	0.336	0.301	0.086	0.078	-0.167	-0.041	0.093	-0.205	1.000				
	Post	-0.224	0.388	0.311	0.093	0.156	-0.161	-0.029	0.098	-0.230					
SO ₄ ²⁻	Pre	-0.183	0.339	0.299	0.220	0.286	-0.134	0.334	0.184	-0.261	0.007	1.000			
	Post	-0.051	0.360	0.310	0.225	0.223	-0.132	0.314	0.183	-0.268	0.012				
Cl ⁻	Pre	0.026	0.298	0.216	0.144	0.160	-0.129	0.240	0.121	0.012	-0.305	0.442	1.000		
	Post	0.219	0.315	0.236	0.164	0.136	-0.126	0.201	0.143	0.034	-0.307	0.493			
NO ₃ ⁻	Pre	0.225	0.119	0.078	-0.007	0.004	-0.251	0.082	-0.025	-0.138	0.119	0.323	0.471	1.000	
	Post	0.273	0.130	0.083	-0.007	0.038	-0.252	0.022	-0.015	-0.114	0.111	0.334	0.484		
F ⁻	Pre	-0.014	-0.201	-0.204	-0.181	-0.278	-0.341	-0.187	-0.178	0.156	0.296	-0.127	-0.243	0.128	1.000
	Post	0.101	-0.160	-0.177	-0.205	-0.218	-0.293	-0.164	-0.180	0.129	0.311	-0.132	-0.245	0.096	

*The top value in each row represents PRM season and bottom values POM season



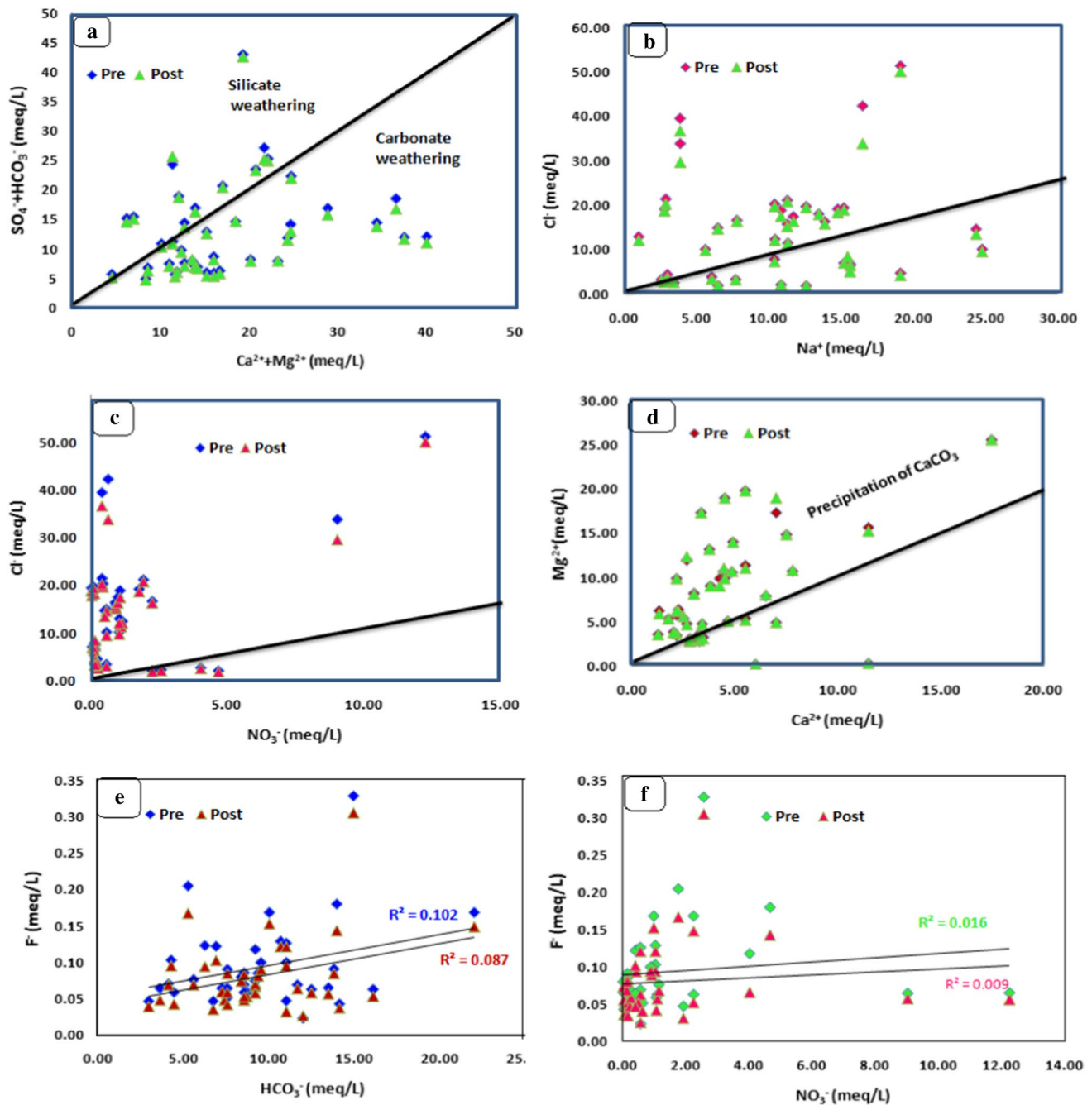


Fig. 5 Scatter diagram showing **A** ($\text{Ca}^{2+} + \text{Mg}^{2+}$) versus ($\text{SO}_4^{2-} + \text{HCO}_3^-$) **B** Na^+ versus Cl^- **C** NO_3^- versus Cl^- **D** Ca^{2+} versus Mg^{2+} **E** HCO_3^- versus F^- **F** NO_3^- versus F^-

(Fig. 5A) demonstrates that the most of the samples are below the equiline, with only a few along the equiline and above. This suggests that carbonate weathering is the primary source of calcium ion in groundwater, while just a

few samples above the equiline suggest silicate weathering (Subbarao et al. 2018; Lakshmanan et al. 2003; Barzegar et al. 2016). Na^+ versus Cl^- almost all groundwater samples fall above the 1:1 equiline (Fig. 5B), suggesting no

halite dissolution. Chloride is comparatively higher than sodium, indicating the absence of silicate weathering (Lakshmanan et al. 2003). Improper domestic sewerage disposal, agricultural influence, animal waste, and excessive use of nitrogenous fertilizers for crop produce could be the main source of higher NO_3^- concentrations in groundwater, as nitrate increases with chloride in many samples (Fig. 5C), both of which are generated by human and animal waste. Because some samples show scattered distribution in the NO_3^- versus Cl^- plot, the elevated nitrate could be attributed to excessive application of both organic and inorganic fertilizers. Furthermore, in the plot of Mg^{2+} versus Ca^{2+} (Fig. 5D), majority of groundwater samples decrease toward Mg^{2+} concentration more than Ca^{2+} content, which is related to CaCO_3 precipitation. According to Datta and Tyagi (1996), the production of calcareous (CaCO_3) concretions implies a detailed level of evaporation due to a semiarid climate, and the presence of CaCO_3 concretions in the soil zone of the current study region confirms this idea. As a result, these findings clearly point to mineral weathering and dissolution, ion exchange, and evaporation as the key controlling elements of groundwater chemical quality. In most cases, high fluoride in groundwater is linked to high bicarbonate ions (Fig. 5E) and in certain circumstances high nitrate ions (Fig. 5F) (Handa 1975; Lakshmanan et al. 2003; Reddy et al. 2010).

Factor analysis

In this study, the first five principle components (PCs) were used to explain 77.54% of the total variance in the PRM season and 77.34% of the total variance in the POM season for both seasons. In PRM season dataset, 32.96% of variances were explained by PC1, 14.01% by PC2, 12.37% by PC3, 9.49% by PC4 and 8.713% by PC5, respectively, and in POM season 27.78% variances are explained by PC1, 15.55% by PC2, 15.07% by PC3, 11.45% by PC4 and 9.50% by PC5, respectively (Table 7). Liu et al. (2003) categorize factor loadings as “strong,” “moderate,” and “weak,” with absolute loading values of > 0.75 , $0.75-0.50$, and $0.50-0.30$, respectively. This categorization was applied to decipher the data in this investigation.

Factor 1 is strongly loaded with EC, TDS, TH, Ca^{2+} and Mg^{2+} during PRM season, and factor 1 is strongly loaded with EC, TDS, TH, Mg^{2+} in POM season. These high loads (EC, TDS, TH, Mg^{2+} , and Ca^{2+}) show the natural hydrogeochemical development of groundwater caused by water–rock interaction that could be explained by chemical weathering dissolving rocks and minerals in sediments. Because clays in the Narji formation and limestones are widely distributed in the research area, EC, TDS, TH, Mg^{2+} , and Ca^{2+} have high values in groundwater. Mg^{2+} ions will rise whenever groundwater interacts with dolomite limestones. The presence of many ions and their compounds resulted in large

Table 7 R-mode factor analysis on the chemical parameters for PRM and POM seasons

Variables	PRM					POM				
	PC1	PC2	PC3	PC4	PC5	PC1	PC2	PC3	PC4	PC5
pH	-0.167	0.039	-0.038	-0.052	0.93	-0.704	0.404	0.079	-0.064	0.067
EC	0.904	0.15	0.029	-0.098	-0.19	0.763	0.27	0.441	0.207	-0.09
TDS	0.963	0.071	-0.012	-0.101	-0.075	0.757	0.206	0.553	0.139	-0.092
TH	0.906	0.042	-0.001	0.096	-0.048	0.928	0.082	0.2	-0.087	0.042
Na^+	0.688	0.022	-0.316	0.023	-0.11	0.349	0.129	0.826	0.081	0.063
K^+	0.343	-0.401	-0.588	0.151	0.041	0.03	-0.325	0.758	-0.284	0.14
Ca^{2+}	0.75	0.168	-0.206	-0.188	0.271	0.479	0.221	0.456	-0.145	-0.236
Mg^{2+}	0.856	0.017	0.021	0.143	-0.122	0.936	0.048	0.114	-0.068	0.113
CO_3^{2-}	0.062	-0.088	0.004	0.924	-0.077	0.076	-0.05	0.092	-0.048	0.936
HCO_3^-	0.271	-0.251	0.678	-0.403	-0.113	0.188	-0.097	0.102	0.847	-0.296
SO_4^{2-}	0.239	0.659	-0.085	-0.351	-0.299	0.249	0.62	0.038	-0.094	-0.4
Cl^-	0.143	0.859	-0.23	0.119	0.014	0.12	0.834	0.021	-0.338	0.073
NO_3^-	0.054	0.704	0.312	-0.074	0.316	-0.091	0.792	-0.028	0.242	-0.011
F^-	-0.167	-0.057	0.79	0.241	0.033	-0.152	0.015	-0.251	0.716	0.295
Eigen value	4.615	1.961	1.731	1.329	1.22	3.89	2.177	2.109	1.601	1.33
% of Variance	32.961	14.01	12.367	9.491	8.713	27.783	15.551	15.066	11.436	9.502
Cumulative %	32.961	46.972	59.339	68.83	77.542	27.783	43.334	58.4	69.837	79.339



Table 8 Classification of water quality based on WQI

Class	WQI value	Water quality status	PRM (2019-Apr)		POM (2018-Sep)	
			Sample numbers	% of samples	Sample numbers	% of samples
I	< 50	Excellent	Nil	Nil	Nil	Nil
II	51–100	Good	23–28,32–34,36,40	27.5	12,14,22–28,32–34, 36,40	35
III	101–200	Poor water	1,2,4,6,8–22,30,35,37–39	60	1–4,6–11,13,15–21, 29–31,35,37–39	62.5
IV	201–300	Very poor water	3,7,29,31	10	5	2.5
V	> 300	Water unsuitable for drinking	5	2.5	Nil	Nil

EC loadings. In both seasons, Factor 2 displays a significant positive loading with Cl^- and a moderate positive loading with SO_4^{2-} and NO_3^- . Factor 2 represents the contribution of agriculture-related nonpoint origin pollution. Factor 3 supports the concept of natural weathering origin due to high fluoride loading and moderate HCO_3^- ion loading with alkali and alkaline earth metals. HCO_3^- can be produced by the carbonate dissolution of minerals, the release of CO_2 from the soil, or the bacterial decomposition of organic contaminants. Factor 4 has strong positive loading on CO_3^{2-} in PRM and POM season, respectively. CO_3^{2-} exhibits substantial positive factor 4 loadings, indicating that it is produced by weathering of carbonates. The pH fluctuation in the aquifer system is significantly loaded in the fifth factor, indicating that groundwater chemistry is governed by it (Table 7). The positive pH loadings suggested that pH fluctuations in the study area regulate major ion concentration.

Drinking water quality index

Water quality index (WQI) was used to measure the suitability of groundwater of Yerraguntla Mandal for drinking purpose. The parameters that will make up the index are chosen based on a number of variables, including the index's purpose, the parameter's importance, and the data's availability (Suvana et al. 2019a, b). The ground water quality is categorized depending on WQI classification, as (I) excellent, (WQI is < 50); (II) good (51–100); (III) poor water (101–200); (IV) very poor water (201–300); (V) unsuitable for drinking, when WQI is > 300 (Table 8). The WQI values of groundwater samples varied from 76.8 to 310.6 with an average of 124.4 during the PRM season, whereas they varied from 55.7 to 287.9 with an average of 139 during POM season. According to WQI classification, 27.5% and 35% of the groundwater samples belong to good water quality; 60%

and 62.5% groundwater samples are of poor water quality, and 10% and 2.5% of the samples are grouped as very poor water quality during PRM and POM seasons, respectively (Table 8).

Spatial distribution of physico-chemical parameters

The application of a geographic information system (GIS) to generate spatial distribution maps, that are point of indicating of suitable and unsuitable regions, is common (Sheikh et al. 2017; Sudhakar and Narsimha 2013; Narsimha and Rajitha 2018; Panaskar et al. 2016; Amiri et al. 2015).

The spatial distribution maps of all physico-chemical parameters are given in Figs. 6, 7 and 8. The pH spatial distribution map depicts that the research area is mostly alkaline. During both seasons, higher concentrations of EC, TDS, and TH can be found in isolated patches in the eastern, north central, and western parts of the research zone. It could also be due to agricultural activities and the interplay of rock and water. Natural sources, sewage, urban run-off, and industrial wastes all contribute to TDS in ground water. Because of the weathering of silicate minerals and sedimentary rocks like limestone and shale, the largest concentrations of Na^+ and K^+ are noticed in the western, north-central, and eastern parts of the research region. Ca^{2+} levels are higher in the north-central and northeastern parts of this region, while Mg^{2+} concentrations are highest in the southwestern and north-central parts. The main sources of elevated Mg^{2+} and Ca^{2+} concentrations are due to the presence of ferromagnesium minerals in basement rocks and the dumping of waste fluids on the ground. Due to decay of organic waste and root respiration that adds up with rainfall (H_2O) to generate HCO_3^- , the maximum concentration is seen in the southern and isolated segments in the southwestern and northern portions of the present region. Sulfates, chloride,



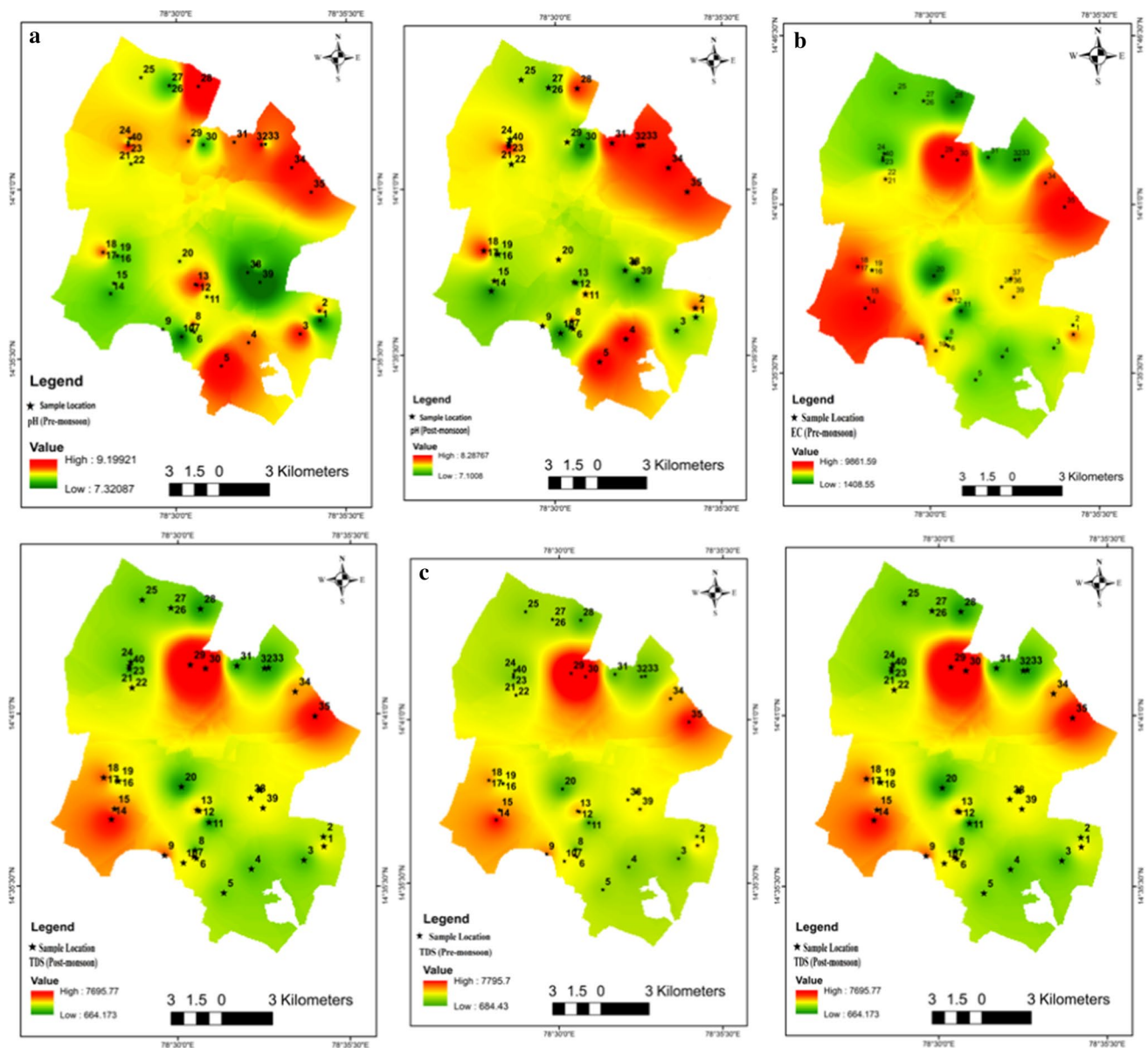


Fig. 6 Spatial distribution map of **a** pH **b** EC **c** TDS during the PRM and POM seasons

and nitrates have an impact on the eastern and northeastern parts of the research region during both seasons, according to distribution maps. Anthropogenic mechanisms (irrigation return flows and poor drainage conditions) are the primary source of SO_4^{2-} , Cl^- , and NO_3^- in groundwater, whereas clays generated by highly weathered rocks are the secondary origin. The ion exchange reaction mechanism, water–rock

interaction, and factors affecting the solubility of minerals in the research region cause F^- pollution in the southern and northeastern parts of the present region. Spatial distribution of water quality index revealed that higher values were observed in southern half of the research region as shown in Fig. 9.



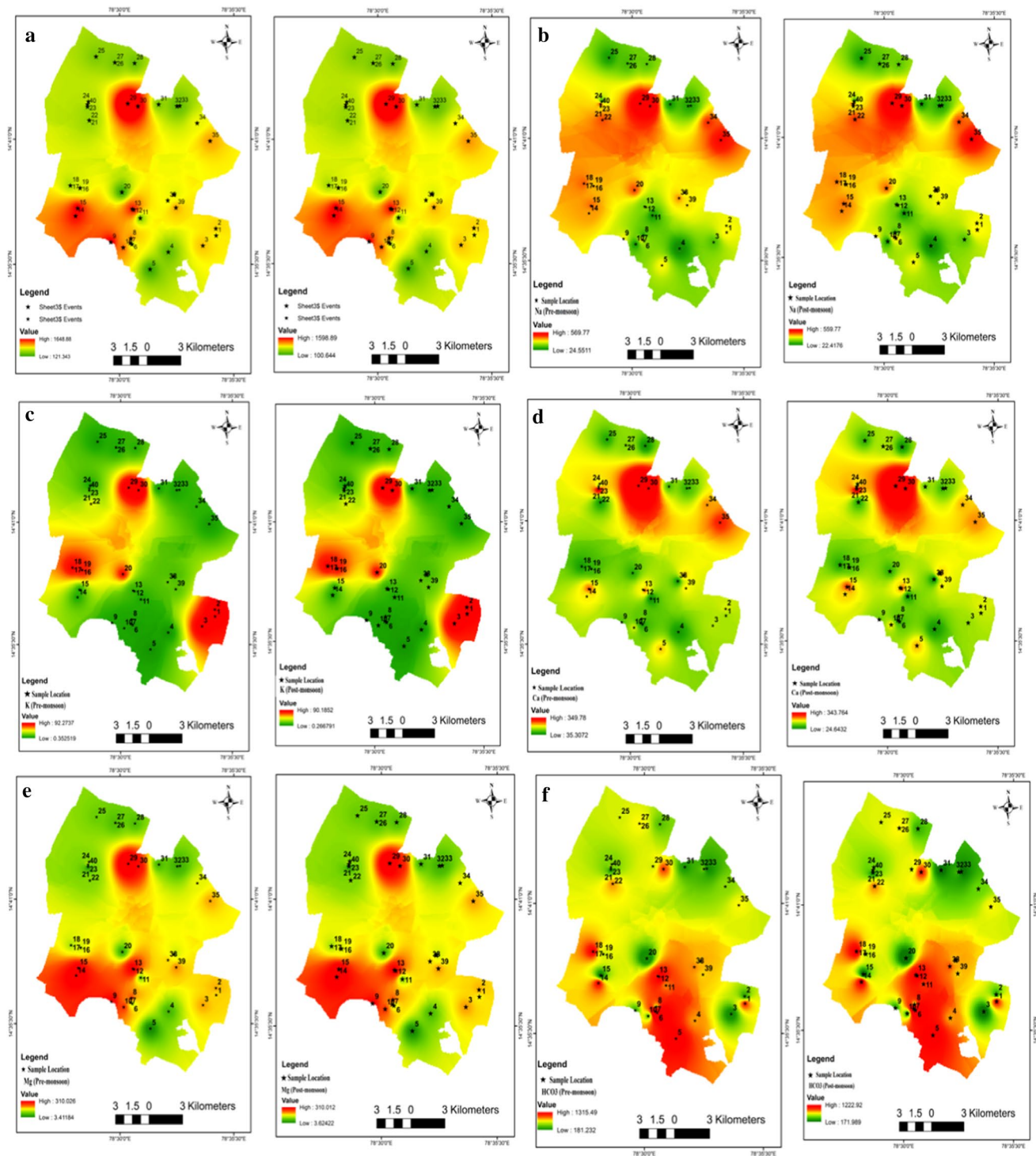


Fig. 7 Spatial distribution map of a TH b Na^+ c K^+ d Ca^{2+} e Mg^{2+} f HCO_3^- during the PRM and POM seasons

Conclusion

This study was mainly intended to study the assessment of groundwater quality for drinking purpose with water quality studies and GIS. Alkaline and hard groundwater has been

discovered. The following are the major cation and anion constituents of water: $\text{Na}^+ > \text{Mg}^{2+} > \text{Ca}^{2+} > \text{K}^+$ for cations and $\text{HCO}_3^- > \text{Cl}^- > \text{SO}_4^{2-} > \text{NO}_3^- > \text{F}^-$ for anions. The analytical data show that Na^+ , HCO_3^- , Cl^- , SO_4^{2-} , NO_3^- , and F^- levels in this location exceed the acceptable limits set by



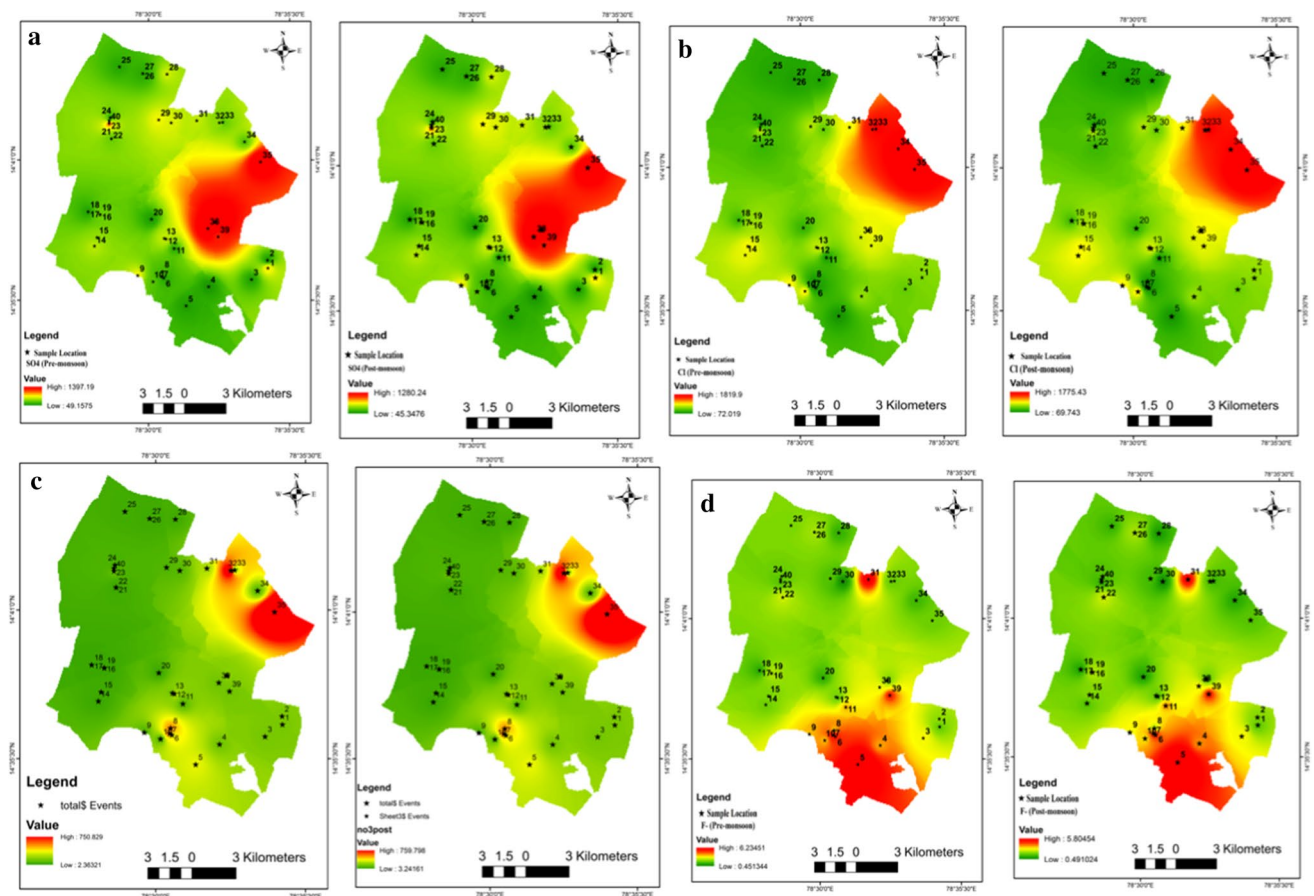


Fig. 8 Spatial distribution map of **a** SO_4^{2-} **b** Cl^- **c** NO_3^- **d** F^- during the PRM and POM seasons

the WHO (2011). The statistical studies were used to better understand the changes in groundwater quality. A bivariate method for describing the degree of relationship among two hydrochemical parameters in PRM and POM seasons is a correlation analysis for EC, TDS, TH, Mg^{2+} , Na^+ , Ca^{2+} , and major ions. Correlation and interpretation in this area signifies that these ions are more probably the outcome of ion exchange mechanism in addition to silicate weathering. Nitrate and chloride levels are positively associated in the research area, indicating contamination from agricultural

and animal waste. 72.5% and 65% of the groundwater samples showed poor and very poor water quality status during the PRM and POM seasons, and overall quality of groundwater is not potable for drinking purpose. Spatial Distribution maps revealed that southern part and small isolated patches of northeastern part are highly affected. The three key components acting in this investigation were found to be related to substantial ion chemical variations in groundwater. Rock–water interaction, chemical weathering, and anthropogenic effects are the main elements that explain the varied loadings.



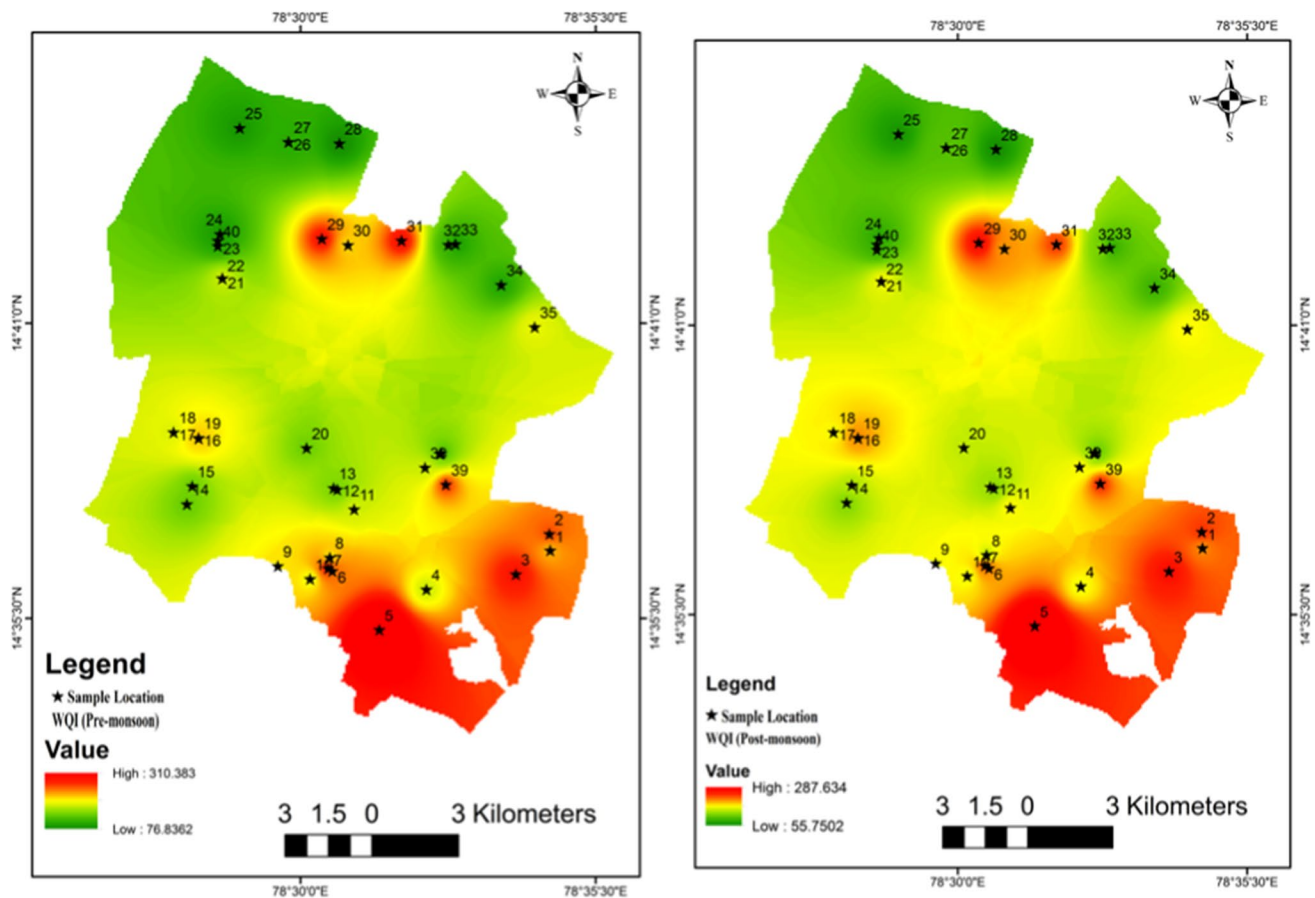


Fig. 9 Spatial distribution map of WQI during the PRM and POM seasons

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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